

Alexander Rubin

(917)-635-3895 | alexrubin345@gmail.com | www.linkedin.com/in/alexander-rubin-engineering

Mechanical Engineering Graduate from the University of Vermont (May 2026) with hands-on experience in CAD design, embedded systems, and prototyping. Proven ability to deliver engineering solutions under real-world constraints through academic projects and field maintenance roles. Seeking an entry-level mechanical engineering position to apply skills in SolidWorks, FEA, and systems integration.

EDUCATION

University of Vermont (UVM), Burlington, VT Graduated: May 2026
Bachelor of Science in Mechanical Engineering GPA: 3.66/4.0

- Related Courses: Applied Thermodynamics, System Dynamics, Fluid Mechanics, Mechanics of Solids, Heat Transfer, Mechatronics, Compressible Flow, Graduate Computational Solids Engineering, Energy Systems Engineering, Computational Fluid Dynamics

RELEVANT PROJECTS

Wearable Wireless Temperature Sensor for Small Aquatic Animals, Burlington, VT August 2025 – May 2026
Course Name: Senior Experience in Engineering Design

- Created a device for the purpose of reading temperature underwater at 30 minute intervals with wirelessly retrievable data.
- Developed and calibrated temperature sensing and conversion algorithms in degrees Celsius, achieving $\pm 0.5^{\circ}\text{C}$ accuracy through temperature chamber testing within the allowable operating conditions.
- Designed a fully integrated circuit (microcontroller, temperature sensor, RFID chip, antenna) enabling contactless data logging and wireless transmission to mobile devices, reducing device footprint to meet sub-0.8g mass constraints.

Autonomous Robotics Competition, Burlington, VT April 2025 – May 2025
Course Name: Mechatronics

- Programmed autonomous navigation logic in Arduino/C++ integrating ultrasonic distance sensing for collision avoidance and object interaction.
- Designed and prototyped an adjustable arm mechanism in SolidWorks that increased task versatility and expanded the robot's effective workspace.
- Integrated sensing and state-machine autonomy to successfully complete 100% of competition tasks under time and constraint requirements.

RS-25 Rocket Engine Atmospheric Performance, Burlington, VT April 2026
Course Name: Computational Fluid Dynamics

- Built and designed accurate RS-25 Rocket Engines in Fluent for 3 different expansion ratios.
- Used real flight data from the Artemis II mission to model a transient pressure model in Fluent to study plume expansion and efficiency while travelling through the atmosphere.
- True boundary and chamber conditions imposed onto the models to show accurate real time flow development within and nozzle and once exposed to real time atmospheric conditions.

RELATED EXPERIENCES

Camp Seneca Lake, Remote August 2025 – May 2026
Property Management Intern

- Modeled a major structural upgrade in SolidWorks to transition from fabric tents to cabin-style shelters, improving camper safety under severe weather conditions.
- Produced detailed Revit blueprints for 20+ property structures, supporting long-term maintenance planning and future infrastructure upgrades.
- Conducted vendor and material research that reduced project preparation time and sourcing effort, saving the property manager over 60+ hours.
- Analyzed utility lifespans and expenses across the property using Excel, identifying patterns that informed a preventative-maintenance schedule and will reduce emergency repair incidents.

Camp Seneca Lake, Penn Yan, NY June – August 2025, 2026
Utility Program Area Leader

- Diagnosed and repaired 20+ mechanical and electrical systems (plumbing, sound systems, lighting circuits) using wiring schematics and datasheets, restoring full functionality and minimizing downtime.
- Served as the primary maintainer for key plumbing infrastructure (sinks, showers, toilets), reducing system downtime through prompt root-cause troubleshooting and repair.
- Conducted daily water-quality testing (pH, bromine, turbidity) to ensure compliance with Department of Health requirements, contributing to successful inspections.

TECHNICAL SKILLS

Technical Skills: SolidWorks, ANSYS Fluent, Revit, AutoCAD, ABAQUS/FEA, Python, Arduino, Excel, LaTeX, C++
Certifications: ABAQUS FEA Essentials, ABAQUS FEA Advanced Training